

EXPLORATORY DATA ANALYSIS (EDA BASICS)

ASSIGNMENT

House Price Analysis Assignment

Dataset: House_Price_India Dataset

Objective

The objective of this assignment is to analyze the House Price India dataset using Python and Pandas.

The analysis includes:

- Univariate Analysis
- Bivariate Analysis
- Business Insights and Interpretation

Step 1: Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

Step 2: Load the Dataset

```
# Load dataset
```

```
df = pd.read_csv('House Price India.csv')
```

```
# Display first 5 rows
```

```
print(df.head())
```

Interpretation

The dataset is successfully loaded and contains information about house prices, bedrooms, bathrooms, living area, location, floors, and other property-related features.

Step 3: Basic Dataset Checks

Dataset Information

```
print(df.info())
```

Statistical Summary

```
print(df.describe())
```

Interpretation

- The dataset contains numerical and categorical information related to houses.

- Statistical summary helps understand average values, minimum values, maximum values, and spread of the data.
- These checks are important before performing analysis.

Step 4: Check Missing Values

```
print(df.isnull().sum())
```

Handling Missing Values

```
# Remove missing values if any
```

```
df = df.dropna()
```

Interpretation

There are no major missing values in the dataset. The dataset is clean and ready for analysis.

Question 1: What is the average, median, and standard deviation of house prices?

```
= average_price = df['Price'].mean()
```

```
median_price = df['Price'].median()
```

```
std_price = df['Price'].std()
```

```
print('Average Price:', average_price)
```

```
print('Median Price:', median_price)
```

```
print('Standard Deviation:', std_price)
```

Answer:

- Average House Price = 538932.22
- Median House Price = 450000
- Standard Deviation = 367532.38

Interpretation

- The average house price is around 538,932.
- The median price is lower than the average, which indicates that some very expensive houses increase the average.
- The high standard deviation shows that house prices vary significantly.

Question 2: Which number of bedrooms is most common?

```
= common_bedrooms = df['number of bedrooms'].mode()[0]
```

```
print('Most Common Bedrooms:', common_bedrooms)
```

Answer:

Most common number of bedrooms = 3

Interpretation

Three-bedroom houses are the most common properties in the dataset, which indicates higher demand and availability for medium-sized family houses.

Question 3: Check if the price data is skewed.

```
= price_skewness = df['Price'].skew()

print('Skewness of Price:', price_skewness)
```

Answer:

Skewness of Price = 4.27

Interpretation

Since the skewness value is greater than 0, the price data is positively skewed. This means:

- Most houses are moderately priced.
- A few houses have extremely high prices.
- The distribution is not symmetrical.

Question 4: What is the average price for each number of bedrooms?

```
= avg_price_bedroom = df.groupby('number of bedrooms')['Price'].mean()

print(avg_price_bedroom)
```

Answer: (Sample)**Bedrooms Average Price**

1	317642
2	401372
3	466000
4	635420
5	786599

Interpretation

- Average price increases as the number of bedrooms increases.
- Larger houses generally cost more because they provide more living space and facilities.

Question 5: What is the relationship between living area and price? (Use: Area without Basement).

= (Use: Area of the house excluding basement)

```
correlation = df['Area of the house(excluding basement)'].corr(df['Price'])
```

```
print('Correlation:', correlation)
```

Answer:

Correlation between area and price = 0.615

Interpretation

There is a positive relationship between house area and price. This means:

- Houses with larger living areas usually have higher prices.
 - The relationship is moderately strong.
-

Question 6: Identify any anomalies where houses have high prices but low areas.

```
= area_q1 = df['Area of the house(excluding basement)'].quantile(0.25)
```

```
price_q3 = df['Price'].quantile(0.75)
```

```
anomalies = df[(df['Price'] > price_q3) &
```

```
(df['Area of the house(excluding basement)'] < area_q1)]
```

```
print(anomalies[['Price', 'Area of the house(excluding basement)', 'number of bedrooms']].head())
```

Answer:

Some houses have high prices despite smaller areas.

Interpretation

Possible reasons for these anomalies:

- Premium locations
- Luxury interiors
- Waterfront view
- Better neighborhood facilities
- High market demand in that area

Question 7: Compare average price based on number of floors and houses with or without waterfront.

Answer: `floor_waterfront = df.groupby(['number of floors', 'waterfront present'])['Price'].mean()
print(floor_waterfront)`

Sample Result

Floors Waterfront Average Price

1.0	0	432372
1.0	1	1211080
1.5	0	547616
2.0	1	1300000

Interpretation

- Houses with waterfront presence are much more expensive.
- Multi-floor houses also tend to have higher prices.
- Waterfront is one of the strongest factors affecting house prices.

Question 8: Identify the minimum and maximum house price. What does this indicate?

```
= minimum_price = df['Price'].min()  
maximum_price = df['Price'].max()  
  
print('Minimum Price:', minimum_price)  
print('Maximum Price:', maximum_price)
```

Answer:

- Minimum Price = 78000
- Maximum Price = 7700000

Interpretation

- The dataset contains both affordable and luxury houses.
- Very expensive properties create a large gap between minimum and maximum values.
- This shows a wide variety of housing categories in the market.

Question 9: Which location (zipcode/are) has the highest average price?

```
= highest_location = df.groupby('Postal Code')['Price'].mean().idxmax()
```

```
highest_avg_price = df.groupby('Postal Code')['Price'].mean().max()
```

```
print('Postal Code:', highest_location)
```

```
print('Highest Average Price:', highest_avg_price)
```

Answer:

- Postal Code with Highest Average Price = 122071
- Highest Average Price = 2348310.81

Interpretation

This postal code represents a premium area where house prices are significantly higher than other locations. Possible reasons include:

- Better infrastructure
- Luxury properties
- Prime location
- High demand

Question 10: Write at least 5 insights from your analysis.

Answer: Insight 1

Three-bedroom houses are the most common properties in the dataset.

Insight 2

House prices are positively skewed because a few luxury houses have extremely high prices.

Insight 3

Larger living areas generally result in higher house prices.

Insight 4

Waterfront properties are much more expensive compared to non-waterfront houses.

Insight 5

Location plays a major role in determining house prices.

Insight 6

Multi-floor houses usually have higher prices due to larger usable space.

Insight 7

Some small-area houses still have very high prices because of premium locations and luxury features.